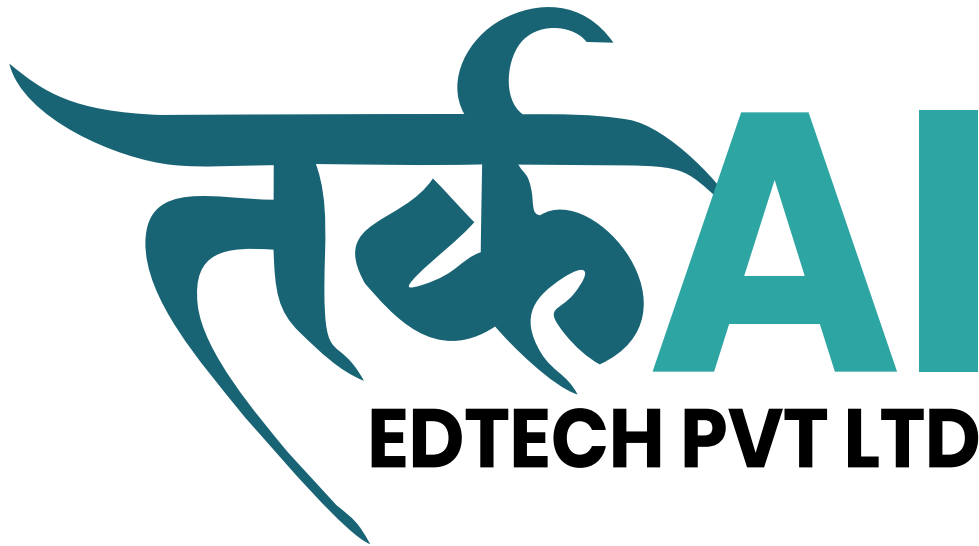


ENGINEERING THE FUTURE OF INTELLIGENCE



“Where intelligence meets education”

AI/ML

DATA SCIENCE

CLIMATE TECH

FOUNDATION

About

TarkAI EdTech Pvt. Ltd.

Founded in 2026 and headquartered in Surat, Gujarat, TarkAI EdTech Pvt. Ltd. is a future-focused education company built on the belief that learning should go beyond certificates and create meaningful impact.

TarkAI was established to address gaps in modern education, where tools are often prioritized over deep understanding, problem-solving, and real-world readiness. We focus on conceptual clarity, practical application, and long-term career value.

We offer structured programs in AI & ML, Data Science, Climate Technology, and Foundation learning for students, professionals, and career switchers through a founder-led, mentor-driven approach. Our philosophy follows Why first, then How, moving from concept -> code -> case studies, with small batches and hands-on projects that build skills, confidence, and career readiness.



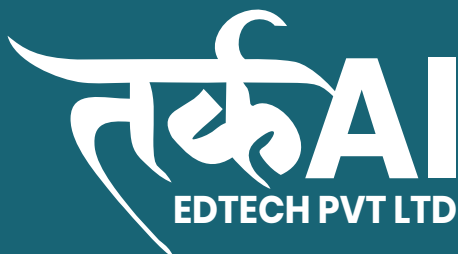
**AI/ML
Architect**

**Data
Science**

**Strategic
Analytics**

**Climate
Analytics**

**AI
Foundation**



Our Vision

To build a future-focused EdTech ecosystem that nurtures curiosity, critical thinking and real-world problem solving, empowering learners to become skilled professionals, ethical innovators, and confident leaders in Artificial Intelligence, Data Science, and Climate Technology.

We envision education that goes beyond certificates—education that creates impact, responsibility, and long-term value for individuals and society.

Our Mission

To deliver industry-relevant, concept-driven education beyond rote learning

To bridge the gap between academic knowledge and real-world application

To mentor learners with a founder-led, hands-on approach

To create future-ready professionals and innovators across AI, Data Science, and Climate Tech

Future Builders



Sahil Rafaliya
Founder & CEO

Sahil Rafaliya is a **Data science professional** with a strong specialization in Climate Technology and AI-driven analytics. He holds an M.Sc. in Data Science (**Climate Data Analytics specialization**) from **Indian Institute of Information Technology, Lucknow**.

With core expertise in **Data Science, AI/ML, and Climate Tech**, he brings a rare blend of technology and sustainability-focused insight to education. Sahil is also the **founder of TrustCarbon and ForITus Solutions**, reflecting his entrepreneurial mindset and commitment to building impactful, future-oriented solutions.

Kashish Nagar is an **AI and Machine Learning specialist** with a strong academic and mentoring background. She holds an **M.Sc. in Artificial Intelligence & Machine Learning** from **Indian Institute of Information Technology, Lucknow**.

Her core expertise spans **Data Science, AI/ML, MLOps, and Teaching & Mentorship**, with a strong focus on building industry-aligned, production-ready skillsets. She is also a **Co-Founder of ForITus Solutions**, contributing to product development, technical architecture, and academic direction.



Kashish Nagar
Co-Founder & President



AI/ML Architect Program

Full-Stack Artificial Intelligence Engineering

Data Science & Strategic Analytics

From Raw Data to Business Intelligence



Future Founders : AI Foundation

Digital Literacy & Coding for the Next Generation

Green Intelligence : Climate Analytics

Specialization in Carbon Markets & ESG Data



AI/ML Architect Program

Full-Stack Artificial Intelligence Engineering

Module 1 – Database Management Systems

- Introduction to DBMS and Data Models
- DBMS Architecture (1-tier, 2-tier, 3-tier)
- Entity Relationship (ER) Modeling
- Relational Model and Keys
- SQL Fundamentals (CRUD Operations)
- Joins, Subqueries, Aggregations
- Normalization (1NF, 2NF, 3NF, BCNF)
- Transactions and ACID Properties
- Indexing and Query Optimization
- Introduction to NoSQL Databases

Outcome: Learners will design efficient databases and write optimized SQL queries.

Module 2 – Python Programming for AI

- Python Syntax and Core Concepts
- Data Structures (List, Tuple, Set, Dictionary)
- Control Flow and Functions
- Object-Oriented Programming (OOP)
- File Handling and Exception Handling
- NumPy for Numerical Computing
- Pandas for Data Analysis
- Data Visualization (Matplotlib, Seaborn)
- Introduction to Scikit-Learn

Outcome: Students gain strong Python foundations for data analysis and ML workflows.

Module 3 – Mathematics & Statistics for AI

- Linear Algebra (Vectors, Matrices, Dot Products)
- Matrix Operations and Intuition
- Probability Fundamentals
- Random Variables and Distributions
- Mean, Variance, Standard Deviation
- Correlation and Covariance
- Bayes Theorem
- Statistical Intuition for ML Models

Outcome: Learners build mathematical intuition required to understand ML algorithms.

Module 4 – Machine Learning

- Types of ML (Supervised, Unsupervised, Reinforcement)
- Bias-Variance Tradeoff, Overfitting
- Regression & Classification Algorithms
- K-Nearest Neighbors (KNN), Naive Bayes
- Decision Trees and Random Forest
- Gradient Boosting & XGBoost
- Clustering (K-Means, Hierarchical)
- Dimensionality Reduction (PCA)
- Model Evaluation Metrics

Outcome: Students build end-to-end ML pipelines and evaluate models effectively.

Module 5 – Deep Learning

- Artificial Neural Networks (ANN)
- Activation Functions and Loss Functions
- Gradient Descent and Backpropagation
- Optimization (SGD, Adam) & Regularization
- Convolutional Neural Networks (CNN)
- CNN Architectures (LeNet, VGG, ResNet)
- Recurrent Neural Networks (RNN)
- LSTM, GRU, and Sequence Modeling

Outcome: Learners understand and implement deep learning models for vision and sequence data.

Module 6 – Computer Vision & NLP

Computer Vision :

- Image Representation & Preprocessing
- Feature Extraction
- CNN-based Vision Pipelines

NLP :

- Text Preprocessing, Tokenization
- Bag of Words, TF-IDF
- Word Embeddings
- Introduction to Transformers

Outcome: Students gain exposure to both vision and language-based AI systems.

Module 7 – LLMs – From Scratch

- Evolution of Language Models
- Transformer Architecture & Self-Attention
- Positional Encoding & Decoder-Only Models (GPT)
- Tokenization Techniques (BPE)
- LLM Training Pipeline (Conceptual)

Outcome: Learners deeply understand how modern LLMs like GPT work internally.

Module 8 – Fine-Tuning LLMs

- Pre-training vs Fine-Tuning
- Instruction Tuning
- Parameter-Efficient Fine-Tuning (PEFT)
- LoRA and QLoRA
- Dataset Preparation, Evaluation and Inference

Outcome: Students can fine-tune open-source LLMs for custom use cases.

Module 9 – Agentic AI

- Introduction to Agentic AI
- Tool-Using AI Agents
- Planning and Reasoning
- Memory-Enabled Agents
- ReAct Framework & Multi-Agent Systems

Outcome: Learners design AI agents capable of decision-making and tool interaction.

Module 10 – Cloud Computing

- Service Models (IaaS, PaaS, SaaS)
- AWS Core Services (EC2, S3, IAM)
- VMs vs Containers
- Docker Fundamentals

Outcome: Students understand how AI systems are deployed and scaled in the cloud.

Module 11: Big Data & MLOps

- Big Data Concepts & Hadoop/Spark Overview
- End-to-End ML Lifecycle
- Experiment Tracking & Model Versioning
- MLOps Tools (MLflow, DVC, Git)

Outcome: Learners build production-ready, maintainable ML systems.

Module 12: Industry Case Studies

Cases :

- Recommendation Systems
- Fraud Detection Systems
- ChatGPT-Style AI Systems
- Healthcare AI Applications
- Autonomous Driving (Perception Layer)

Outcome: Students develop interview-ready explanations and problem-solving skills.

Data Science & Strategic Analytics

From Raw Data to Business Intelligence

Phase 1 : Programming & Data Foundations

Module 1 – Python for Data Science

- Python fundamentals & syntax
- Data types, Control flow, Functions
- OOP, File handling, Exceptions
- NumPy: Arrays & vectorized operations
- Pandas: DataFrames, Cleaning, Merging
- Visualization: Matplotlib, Seaborn

Outcome: Learners can clean, analyze, and visualize data using Python.

Module 2 – Database Management & SQL

- DBMS Intro, Relational databases, ER diagrams
- SQL (SELECT, Joins, Subqueries, GROUP BY)
- Constraints & Indexes
- Intro to NoSQL (MongoDB basics)

Outcome: Learners can extract and manipulate data using SQL.

Phase 2 : Mathematics, Statistics & Analysis

Module 3 – Mathematics for Data Science

- Linear Algebra (Vectors & matrices)
- Calculus (Derivatives, Gradient)
- Distance measures

Outcome: Learners understand math behind models and metrics.

Module 4 – Statistics & Probability

- Descriptive statistics & Probability theory
- Random variables & Distributions
- Bayes' theorem & Sampling techniques
- Hypothesis testing (Z-test, T-test, Chi-square)
- A/B testing

Outcome: Learners can validate insights statistically.

Phase 3: Analysis & Visualization

Module 5 – Exploratory Data Analysis (EDA)

- Data cleaning strategies
- Handling missing values
- Outlier detection
- Correlation analysis & Pattern identification

Module 6 – Data Visualization & BI Tools

- Storytelling with data
- Dashboards & reports
- Power BI / Tableau
- KPI & metric design

Outcome: Learners can present business insights clearly.

Phase 4: Machine Learning for Data Science

Module 7 – Machine Learning Fundamentals

- ML workflow & Preprocessing
- Feature engineering
- Linear Regression & Logistic Regression
- KNN & Naive Bayes
- Model evaluation metrics

Module 8: Advanced ML Models

- Decision Trees & Random Forest
- Gradient Boosting & XGBoost
- Model tuning (Grid Search, Cross-validation)

Outcome: End-to-end ML predictive model project.

Phase 5 : Big Data & Deployment

Module 9 – Big Data Analytics

- Big Data concepts (5Vs)
- Hadoop ecosystem
- Apache Spark basics (DataFrames, SQL)

Module 10 – Cloud & MLOps for Data Science

- AWS basics (S3, EC2)
- Docker basics
- ML lifecycle & Version control (Git)
- Intro to MLflow & DVC

Phase 6: Capstone Projects

Module 11 & 12 – Case Studies & Capstone

- Customer churn analysis
- Sales forecasting
- Fraud detection

Capstone: End-to-end project (Business problem -> ML model -> Dashboard).

Additional Benefits : 1 Month Placement Ready Program

AI/ML Architect Program
Data Science & Strategic Analytics

ATS-Friendly Resume Building

- Resume optimized for MNCs & startups
- Keyword-based, HR-approved formats

Mock Interviews (Technical + HR)

- Real interview environment
- Personalized feedback & improvement plan

Industry Expert Mentorship

- Guidance from professionals working in the same field
- Insights on hiring trends & expectations

Portfolio Building

- Project-based portfolio
- GitHub + live project review

LinkedIn Profile Optimization

- Recruiter-friendly headline & summary
- Job-search & networking strategy

Placement Readiness Support

- Interview question bank
- Salary negotiation tips
- Confidence & communication grooming

Future Founders : AI Foundation

Digital Literacy & Coding for the Next Generation

Module 1: Digital Literacy & Data Basics

- Types of data (numbers, text, images)
- Data in daily life
- AI vs Human thinking

Tools: Google Sheets, ChatGPT demos.

Module 5: Data Analysis & Visualization

- Tables, Mean/Max/Min
- Bar charts, Line charts
- Why visualization matters

Tools: Pandas, Matplotlib.

Module 2: Logical Thinking & Algorithms

- Logic & Step-by-step thinking
- Algorithms (no code)
- Flowcharts & Decision making

Tools: Draw.io, Scratch.

Module 6: Intro to AI & ML

- What is Machine Learning?
- Training vs Testing
- Prediction concept

Tools: Teachable Machine, Scikit-learn demo.

Module 3: Introduction to Coding

- Variables & values
- Conditions (if-else)
- Loops (repeat)

Tools: Scratch (Block), Python basics (Text).

Module 7: AI Tools, Ethics & Careers

- Responsible AI usage
- Bias & fairness
- Career paths in AI/Data

Tools: Canva, ChatGPT.

Module 4: Python & Data Handling

- Python basics
- Working with lists
- Reading simple datasets

Tools: Google Colab, Excel/Sheets.

Module 8: Mini Project

- Student marks analysis
- Survey data insights
- AI image classification demo

Green Intelligence : Climate Analytics

Specialization in Carbon Markets & ESG Data

Part 1: Climate Data Analytics

Module 1: Climate Science Essentials

- Weather vs Climate (data view)
- Climate system & carbon cycle
- Greenhouse gases (CO₂, CH₄, N₂O)
- Global warming potential (GWP)
- Climate indicators

Outcome: Understand climate indicators that directly affect carbon markets.

Module 2: Climate & Emissions Data

- Time-series, Country-level, Sector-wise data
- Sources: IPCC, UNFCCC, World Bank
- Formats: CSV, NetCDF (concept)

Outcome: Learn where emissions and climate data comes from.

Module 3: Data Processing

- Cleaning emissions data
- Handling missing reporting
- Aggregation & Normalization

Hands-on: CO₂ emissions preprocessing.

Module 4: Trends & Analysis

- Time-series analysis
- Emissions growth vs reduction trends
- Correlation (GDP vs emissions)
- Anomaly detection

Tools: Matplotlib, Seaborn.

Part 2: Carbon Markets & Analytics

Module 5: Introduction to Carbon Markets

- Carbon credits vs offsets
- Compliance vs Voluntary markets
- Cap-and-trade (EU ETS)
- India Carbon Market (ICM)
- Carbon pricing & Allowance allocation

Module 6: Carbon Accounting

- Scope 1, Scope 2, Scope 3 emissions
- Emission factors
- Corporate GHG inventories & ESG reporting

Hands-on: Calculate emissions using activity data.

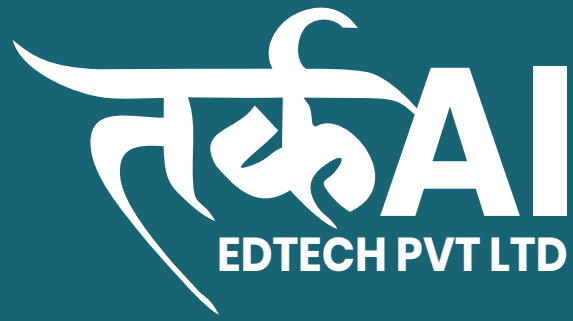
Module 7: MRV Systems

- Measurement, Reporting & Verification (MRV)
- GHG Protocol & ISO 14064
- Verification process & Audit trails
- Baseline vs project emissions

Module 8: Capstone Project

- Country-wise carbon allowance gap analysis
- Carbon price trend analysis
- Corporate Scope 1-2 emissions dashboard

Outcome: Apply data science to carbon market problems.



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